Hivemind control-data map

#	CHECKPOINT	OBSERVE	DELAY	DENY	TAMPER	SELECT
L1: libp2p transport	Bootstrap peer selection lives here and is unexamined.					
1	Bootstrap and transport discovery	untested	untested	untested	untested	untested
L2: Kademlia DHT	One fix, *node_id* derived from the authenticated key, changes every row.					
2	DHT reachability and time check	untested	untested	untested	untested	untested
3	Routing admission, liveness, eviction	untested	untested	untested	**C3.T**	untested
4	Lookup referrals	untested	untested	untested	**C7.T**	untested
5	Record responsibility transfer	untested	untested	untested	untested	untested
6	Record publication and replication	untested	untested	untested	untested	**C6.S**
7	Record lookup and local-view construction	untested	untested	untested	**C7.T**	**C7.S**
8	Record refresh, withdrawal, expiration	untested	untested	untested	**C8.T**	untested
L3: Matchmaking	One fix, signed advertisements, changes rows 11 and 15.					
11	Averaging availability exchange *(also E01)*	**C11.Obs**	**C11.Delay**	**C11.D**	**C11.T**	**C7.S**
12	Candidate priority and leader choice *(local, also E01)*	untested	untested	untested	untested	**C12.S**
13	Join negotiation and admission *(also E01)*	untested	untested	**C13.D**	untested	**C13.S**
14	Group roster distribution	untested	untested	*as 13*	untested	*as 13*
15	Next-group-key derivation *(local)*	untested	untested	untested	untested	**C15.S**
L4: All-reduce	Control data: roles (16), recovery (18), weight and group id (17).					
16	Role and work assignment *(local)*	untested	untested	untested	untested	untested
17	All-reduce exchange *(payload too)*	untested	untested	untested	**C17.T**	untested
18	Session failure and recovery	untested	untested	untested	untested	untested
L5: Optimizer	Progress records *are* signed; L3's advertisements are not.					
9	Training-progress exchange	untested	untested	untested	untested	untested
10	Progress and synchronisation decision *(local)*	untested	untested	untested	untested	untested
L6: State transfer	Donor priority self-declared, tensors installed unvalidated. Replaces a model, not nudges it.					
19	State-donor advertisement and selection	untested	untested	untested	untested	untested
20	State download and installation *(payload too)*	untested	untested	untested	untested	untested
Spanning L2 / L3 / L5						
21	Shutdown and withdrawal	untested	untested	**C21.D**	untested	

vulnerability demonstrated in a live swarm **defence holds only within a bound** **no vulnerability**, prevented with a reason **untested**, nobody has looked blank = n/a